

# 1 Solution — GIAM 3.1

## Problem 12

### 1.1 Problem

Suppose  $a, b, c, d$  are integers with  $a \neq c$ , and  $x$  is a real number satisfying

$$\frac{ax + b}{cx + d} = 1.$$

Show that  $x$  is rational. **Where is the hypothesis  $a \neq c$  used?**

### 1.2 Proof

**Proof.** Since the left side is written as a fraction,  $cx + d \neq 0$ . Multiplying both sides by  $cx + d$ ,

$$ax + b = cx + d.$$

Collecting the  $x$  terms on one side,

$$(a - c)x = d - b.$$

Because  $a \neq c$  we have  $a - c \neq 0$ , so we may divide by it:

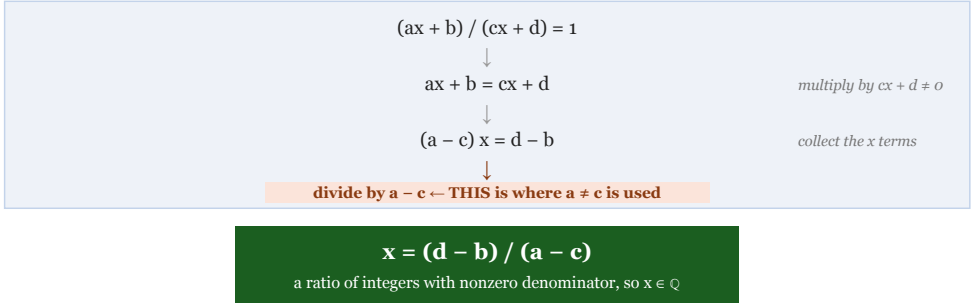
$$x = \frac{d - b}{a - c}.$$

Now  $d - b$  and  $a - c$  are integers ( $\mathbb{Z}$  is closed under subtraction) and  $a - c \neq 0$ . So  $x$  is a ratio of integers with nonzero denominator, hence  $x \in \mathbb{Q}$ . ■

### 1.3 Where the Hypothesis Is Used

$a \neq c$  is used at exactly one step: it licenses dividing by  $a - c$ .

Clear the fraction, collect the  $x$  terms, then divide — and the division is where the hypothesis lives.



Drop the hypothesis and the theorem collapses — in two different ways

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| <p style="text-align: center; color: #8b0000;"><b><math>a = c</math> but <math>b \neq d</math></b></p> <p style="text-align: center;"> <math>(a - c)x = d - b</math> becomes <math>0 = d - b</math><br/>             which is false         </p> <p style="text-align: center; color: #8b0000;"><b>no <math>x</math> satisfies the equation</b></p> <p style="text-align: center; font-size: small;">e.g. <math>a = c = 1, b = 5, d = 7</math></p> | <p style="text-align: center; color: #8b0000;"><b><math>a = c</math> and <math>b = d</math></b></p> <p style="text-align: center;"> <math>(a - c)x = d - b</math> becomes <math>0 = 0</math><br/>             which is true for every <math>x</math> </p> <p style="text-align: center; color: #8b0000;"><b><math>x</math> is not pinned down at all</b></p> <p style="text-align: center; font-size: small;">e.g. <math>a = c = 1, b = d = 0</math></p> |
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**The second case actually produces an irrational solution**

take  $a = c = 1, b = d = 0$ : the equation reads  $x / x = 1$   
 which holds for every  $x \neq 0$  — including  $x = \sqrt{2}$   
**so without  $a \neq c$  the conclusion " $x$  is rational" is simply false**

Figure 1.1: Top: the derivation. From  $(ax \text{ plus } b) \text{ over } (cx \text{ plus } d) \text{ equals } 1$ , multiply by  $cx \text{ plus } d$  which is nonzero to get  $ax \text{ plus } b \text{ equals } cx \text{ plus } d$ ; collect the  $x$  terms to get  $(a \text{ minus } c) \text{ times } x \text{ equals } d \text{ minus } b$ ; then divide by  $a \text{ minus } c$  — this step is highlighted as where the hypothesis  $a \text{ not equal to } c$  is used. A green banner gives the answer:  $x \text{ equals } (d \text{ minus } b) \text{ over } (a \text{ minus } c)$ , a ratio of integers with nonzero denominator, so  $x$  is rational. Below, two panels showing what goes wrong without the hypothesis. First,  $a \text{ equals } c$  but  $b \text{ not equal to } d$ : the equation  $(a \text{ minus } c)x \text{ equals } d \text{ minus } b$  becomes  $0 \text{ equals } d \text{ minus } b$ , which is false, so no  $x$  satisfies the equation — for example  $a \text{ equals } c \text{ equals } 1, b \text{ equals } 5, d \text{ equals } 7$ . Second,  $a \text{ equals } c$  and  $b \text{ equals } d$ : the equation becomes  $0 \text{ equals } 0$ , true for every  $x$ , so  $x$  is not pinned down at all — for example  $a \text{ equals } c \text{ equals } 1, b \text{ equals } d \text{ equals } 0$ . A final panel: the second case actually produces an irrational solution, since with  $a \text{ equals } c \text{ equals } 1$  and  $b \text{ equals } d \text{ equals } 0$  the equation reads  $x \text{ over } x \text{ equals } 1$ , which holds for every nonzero  $x$  including the square root of 2 — so without  $a \text{ not equal to } c$  the conclusion that  $x$  is rational is simply false.

Everything up to  $(a - c)x = d - b$  is valid for *any* integers. The hypothesis is needed only to know that  $a - c$  is a **legal divisor**, and that single step is what converts the

equation into a formula for  $x$ .

## 1.4 Remark — The Hypothesis Is Genuinely Necessary

It is not enough to note where a hypothesis is used; one should check the theorem actually *fails* without it. Dropping  $a \neq c$  splits into two cases, and both are instructive.

**Case  $a = c$ ,  $b \neq d$ .** The equation  $(a - c)x = d - b$  becomes  $0 = d - b$ , which is false. So **no** real  $x$  satisfies the original equation — the statement is vacuously true but says nothing. With  $a = c = 1$ ,  $b = 5$ ,  $d = 7$ , no solutions exist.

**Case  $a = c$ ,  $b = d$ .** Now the equation becomes  $0 = 0$ , true for *every*  $x$ . The original fraction is  $\frac{ax + b}{ax + b} = 1$ , which holds for every  $x$  avoiding the pole. Taking  $a = c = 1$ ,  $b = d = 0$ , the equation reads  $x/x = 1$ , satisfied by

$$x = \sqrt{2},$$

which is **irrational** (Section 1.6). So the conclusion is outright false without the hypothesis.

That second case is the decisive one. A hypothesis whose removal merely makes a theorem unprovable is suspicious; a hypothesis whose removal makes it *false* is doing real work.

## 1.5 Remark — The Hidden Hypothesis $cx + d \neq 0$

There is a second assumption the problem never states: for  $\frac{ax + b}{cx + d}$  to be a number, the denominator must be nonzero. Writing the equation at all presupposes it.

This matters because the first step multiplies both sides by  $cx + d$ . Multiplying an equation by zero is always permitted but destroys information —  $0 = 0$  follows from anything. Here the multiplication is safe precisely because  $cx + d \neq 0$  is guaranteed by the expression being well-formed.

Such assumptions are easy to overlook because they are carried by *notation* rather than stated in words. The habit worth building is to ask, of every expression in a hypothesis, what has to be true for it to make sense at all.

## 1.6 Remark — The Answer Is Already in Lowest Terms? Not Necessarily

The proof produces  $x = \frac{d-b}{a-c}$ , a legitimate ratio of integers. But if one uses the strict Section 1.6 definition of  $\mathbb{Q}$  — requiring  $\gcd$  of numerator and denominator to be 1 — then, exactly as in Problem 3, an extra step is needed: divide both by  $g = \gcd(d-b, a-c)$ .

For instance  $a = 5$ ,  $c = 1$ ,  $b = 1$ ,  $d = 9$  give  $x = \frac{8}{4}$ , which is not reduced. Dividing by  $\gcd(8, 4) = 4$  gives  $x = \frac{2}{1}$ . The value is right either way; only the *representative* needed fixing. As before, whether this step is required depends on which definition of  $\mathbb{Q}$  is in force.

## 1.7 Remark — Every Such Equation Has a Rational Solution

The proof is **constructive**: it does not merely show a rational solution exists, it produces the formula  $x = (d-b)/(a-c)$  explicitly. That is worth noting because the problem only asked to *show  $x$  is rational*, and a non-constructive argument would have sufficed.

The formula also settles more than was asked. Since  $(a-c)x = d-b$  has a *unique* solution when  $a \neq c$ , the equation determines  $x$  completely — there is exactly one real number satisfying it (provided it avoids the pole  $cx + d = 0$ ). So the theorem could be strengthened to " $x$  is the unique rational  $(d-b)/(a-c)$ ," and the proof already establishes it. This is the same "read your own witness" point as Problem 7.

## 1.8 Remark — Why the Equation Being Set to 1 Matters

Setting the fraction equal to 1 is what makes everything collapse so neatly: cross-multiplying gives a **linear** equation, whose solution is a single quotient.

Had the right side been another rational, say  $\frac{ax+b}{cx+d} = \frac{p}{q}$ , cross-multiplying gives  $q(ax+b) = p(cx+d)$ , so  $(qa - pc)x = pd - qb$  and  $x = \frac{pd - qb}{qa - pc}$  — still rational, with the hypothesis now being  $qa \neq pc$ . The argument is unchanged in shape; only the arithmetic thickens. What would genuinely break it is a right side that is *irrational*: setting the fraction equal to  $\sqrt{2}$  gives  $x = \frac{d\sqrt{2} - b}{a - c\sqrt{2}}$ , which is generally irrational. The rationality of  $x$  traces back entirely to every constant in sight being rational.